HACKSPHERE 2026
PHASE 1 PROPOSAL IDEA

Project Title
DanaTerang: Transparent Donation & CSR Disbursement Tracking

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1. Executive Summary

DanaTerang is a transparent donation and CSR disbursement platform that lets funders see how money moves from campaign goals to verified milestones. Instead of relying on a final narrative report, organizers submit milestone evidence such as receipts, photos, and short reports; DanaTerang runs OCR and consistency checks, creates a cryptographic hash of the evidence package, and anchors the milestone record and timestamp in a smart contract. Donors, CSR teams, beneficiaries, and the public can then verify the same audit trail without requiring a token or cryptocurrency transaction.

2. Problem Statement

Donation and CSR programs depend on trust, but the evidence trail is often fragmented across chats, spreadsheets, cloud folders, and post-hoc reports. The Indonesian Philanthropy Association explicitly identifies mismatched use or distribution of donations caused by limited transparency and accountability as a recurring philanthropy ethics problem. A concrete Indonesian case also shows the cost of weak oversight: in January 2023, the South Jakarta District Court convicted former ACT president Ahyudin in a case involving Boeing Community Investment Fund donations; reporting on the judgment stated that Rp117 billion of Rp138.546 billion received was used outside the agreed implementation, while Rp20.564 billion was implemented for the intended beneficiaries. These facts do not mean every charity is problematic; they show why donors need a verifiable, evidence-linked audit trail rather than trust alone.

DanaTerang targets the gap between "a campaign has reported spending" and "an independent observer can verify each disbursement milestone." The product focuses on evidence integrity and traceability—not on replacing legal audits or financial institutions.

Sources: Indonesian Philanthropy Association, "Etika Filantropi Indonesia: Membangun Kepercayaan Publik"; Kompas/ANTARA reporting on the January 2023 South Jakarta District Court judgment in the ACT case. Links: https://filantropi.or.id/en/national-news/etika-filantropi-indonesia/
|https://nasional.kompas.com/read/2023/01/24/17045171/pendiri-act-ahyudin-divonis-35-tahun-penjara
|https://www.antaranews.com/berita/3362724/pendiri-act-ahyudin-divonis-tiga-setengah-tahun-penjara

3. Proposed Solution & Value Proposition

DanaTerang turns each campaign into a sequence of auditable milestones. A campaign owner defines the purpose, target amount, and milestone conditions. When a milestone is completed, the organizer uploads the supporting evidence; the platform extracts receipt data with OCR, checks basic consistency between the declared expense and the document, and assigns an anomaly flag when the evidence is incomplete or unusual. A normalized evidence package is hashed with SHA-256, while the hash, milestone ID, timestamp, and status are written to an EVM-compatible smart contract. The original files remain in conventional storage for privacy and lower cost.

The key value proposition is "verify the evidence trail, not just the promise." Compared with ordinary campaign pages or static reports, DanaTerang provides a public verification layer that makes later alteration detectable. Compared with putting all documents on-chain, it keeps sensitive evidence off-chain and uses blockchain only for the minimum immutable proof needed for auditability. No token, marketplace, or DeFi mechanism is required.

4. Minimum Viable Product (MVP) / Key Features

The Phase 2 build will focus on five features that are directly demonstrable within the 24-hour coding marathon.

1. **Campaign & Milestone Creation —** Organizer creates a campaign, funding purpose, target, and 2–5 disbursement milestones.
2. **Evidence Upload & Hashing —** Receipt/photo/report files are stored with metadata and a SHA-256 evidence hash.
3. **AI-Assisted Receipt Verification —** OCR extracts vendor, date, total, and text; consistency checks flag missing fields, amount mismatches, duplicate evidence, or suspicious date/amount combinations.
4. **On-Chain Milestone Anchoring —** A Solidity contract records campaign ID, milestone ID, evidence hash, status, and timestamp/transaction reference; no token is issued.

5. **Public Audit & Verification Page —** Anyone can inspect milestone status, evidence metadata, AI flags, and on-chain proof, then re-hash a file to confirm integrity.

5. Target Market & Impact Viability

Primary active users are: (1) Indonesian nonprofit and community organizations running small-to-medium donation campaigns; (2) CSR teams that need evidence-linked reporting for funded projects; and (3) donors who want a simple public verification page instead of requesting documents individually. Secondary users include independent verifiers, journalists, community leaders, and partner institutions.

The MVP is designed for one campaign and a small number of milestones, but the architecture scales by separating file storage from the blockchain proof layer. In later versions, DanaTerang can support organization profiles, verifier roles, recurring CSR programs, beneficiary confirmation, exportable audit reports, and integrations with payment or accounting systems. Social impact comes from reducing information asymmetry: organizers gain a standardized reporting workflow, while funders and observers gain a tamper-evident trail that is easier to inspect.

6. Tech Stack & System Architecture

Frontend: Next.js + TypeScript + Tailwind CSS for a responsive web dashboard and public audit page.

Backend: Node.js + Express for campaign, milestone, evidence, verification, and audit APIs.

Database: PostgreSQL for structured campaign/milestone metadata.

Storage: S3-compatible object storage for receipts, photos, and reports; sensitive files are not written directly to the blockchain.

AI/OCR: Tesseract.js for receipt OCR plus lightweight anomaly rules for amount/date/field consistency and duplicate-evidence checks. This keeps the AI path functional and feasible in a 24-hour build.

Web3: Solidity smart contract deployed to an EVM-compatible testnet; ethers.js handles contract interaction. Only the evidence hash and essential milestone proof are anchored on-chain.

Security: SHA-256 evidence hashing, role-based access control, server-side validation, and signed transaction records.

Data flow: User → Next.js frontend → Node.js API → PostgreSQL/Object Storage. Evidence also passes through OCR/anomaly checks. After submission, the backend hashes the evidence package and sends the proof to the smart contract. The public audit page reads both the database metadata and on-chain transaction proof.

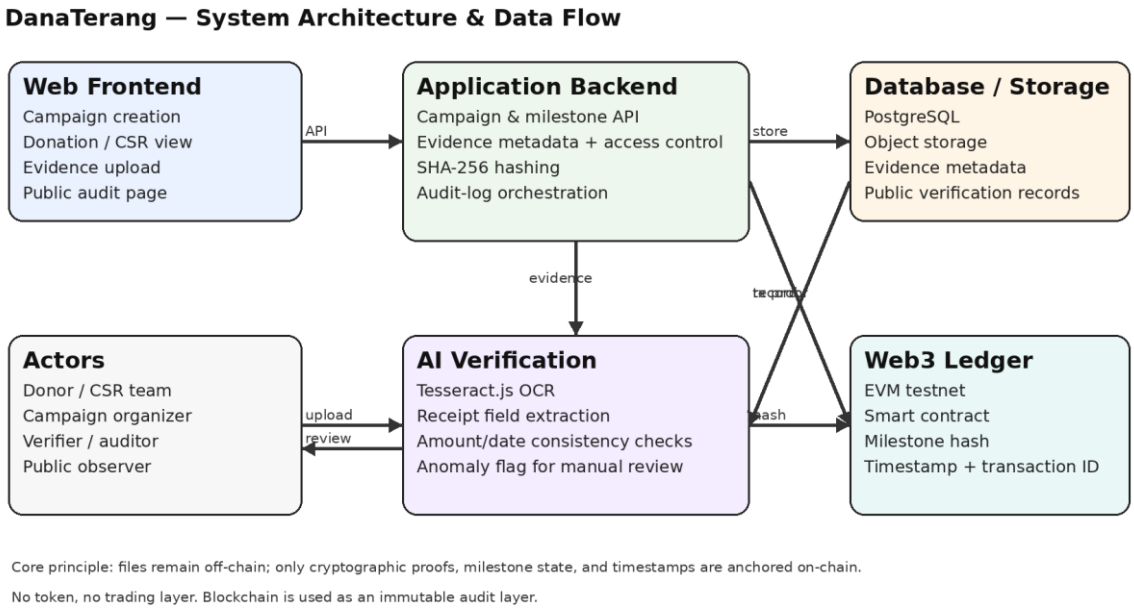


Figure 1. DanaTerang functional data flow: conventional application services handle files and metadata; blockchain stores only tamper-evident proof.

Sources & Scope Note

Problem evidence: Indonesian Philanthropy Association, “Etika Filantropi Indonesia: Membangun Kepercayaan Publik”; Kompas/ANTARA reporting on the January 2023 South Jakarta District Court judgment in the ACT case. DanaTerang is an auditability and evidence-integrity layer. It does not claim to replace statutory audits, financial institutions, legal investigations, or government reporting requirements.